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Supply Chain Scheduling: Conflict and Cooperation in Assembly Systems

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We study conflict and cooperation issues in supply chain manufacturing. Consider an assembly system where suppliers provide parts to a manufacturer. A product cannot be delivered until all its parts have been supplied. The manufacturer performs nonbottleneck operations, for example, outsourced assembly, packaging, and delivery for each product. Two classical scheduling objectives are considered: minimization of the total completion time and of the maximum lateness. We analyze how far from optimal the best schedule for a suppliers' scheduling problem can be for the corresponding manufacturer's problem, and vice versa. To resolve these conflicts, we consider four alternative scenarios for the relative bargaining power of the suppliers and the manufacturer, and in each case describe a practical mechanism for cooperation between the decision makers. Evaluating the cost of conflict and the benefit of cooperation in these scenarios requires the solution of various scheduling problems by the suppliers, the manufacturer, and the overall system. For all these scheduling problems, we provide either an efficient algorithm or a proof of intractability. Moreover, for two problems that we show are intractable, we describe heuristics and analyze their worst case performance or demonstrate asymptotic optimality of their solutions. We demonstrate computationally that the cost saving realized by cooperation between the decision makers is significant in many cases. Extensions of our models to consider bottleneck operations at the manufacturer and transportation times are also developed.

Subject classifications: supply chain scheduling; decision making: cooperation; cost of conflict; cooperation: benefit.

Area of review: Manufacturing, Service, and Supply Chain Operations.

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1. Introduction

One of the most active topics in manufacturing research over the last 10 years has been supply chain management. A supply chain represents all the stages at which value is added to a manufacturing product. These stages include the supply of raw materials and intermediate parts, finished goods manufacture, packaging, transportation, warehousing, and logistics. This paper focuses on the important issues of conflict and cooperation between decision makers in an assembly system.

In an assembly system, there are several suppliers and one manufacturer. Each supplier provides parts to the manufacturer. The manufacturer waits until all the parts for a job have arrived and then initiates a final stage of manufacturing. To focus on conflict and coordination issues instead of constraints faced by the manufacturer, we assume that this final stage is nonbottleneck, however, none of the cooperation mechanisms discussed in the paper rely on this assumption. In nonbottleneck manufacturing, as many products as needed can be processed through the final stage simultaneously. Practical examples of nonbottleneck final manufacturing stages include outsourced assembly,

packaging, and delivery. As is customary in assembly systems, different parts of the same job can be processed simultaneously by different suppliers. The objective is to minimize the overall scheduling cost.

Thomas and Griffin (1996) provide a review and discussion of the literature on supply chain management. They point out that over 11% of the U.S. Gross National Product is devoted to nonmilitary logistics expenditures. Moreover, for many products, logistics expenditures constitute over 30% of the cost of goods sold. In their conclusions, they discuss the need for research that addresses supply chain issues at an *operational* rather than a strategic level, and that uses *deterministic* rather than stochastic models. This paper directly addresses both of these needs. Sarmiento and Nagi (1999) survey the literature of integrated production and distribution models. They point out that recent reductions in inventory levels (Feld 2000, Rajagopalan and Malhotra 2001) create a closer interaction between the stages in a supply chain, which increases the need for cooperation in decision making. They also discuss several possible topics for future research. In view of these two extensive survey papers, we do not discuss the general supply chain management literature in detail.

Within traditional supply chain management research, the literature that is most closely related to this paper is on production and distribution systems with multiple stages. Ow et al. (1988) describe a multiagent, or distributed, scheduling system. However, they do not evaluate the benefits of cooperation in detail. Lee and Chen (2001) consider the integration of transportation constraints, including time and capacity, with scheduling decisions. For a variety of models, they provide either an efficient algorithm or a proof of intractability. There are at least two previous studies of scheduling problems in assembly systems. Lee et al. (1993) consider the minimization of makespan in an assembly system with two suppliers and one manufacturer. They prove that this problem is intractable, provide an enumerative algorithm, and analyze special cases and heuristics. Potts et al. (1995) consider a more general problem with several suppliers and one manufacturer, and describe heuristics with small worst case error bounds. However, none of the above four papers discusses the issues of conflict and cooperation between decision makers which are the main focus of our work.

The first study of supply chain scheduling, by Hall and Potts (2003), evaluates the benefits of cooperative decision making in a supply chain where a supplier makes deliveries to several manufacturers, who may in turn supply several customers. They develop models which minimize the total of scheduling and batch delivery costs. They demonstrate that cooperation between a supplier and a manufacturer may reduce the total cost in this system by at least 20% or as much as 100%, depending upon the scheduling objective being considered. Agnetis et al. (2006) study models for resequencing jobs, using limited storage buffer capacity, between a supplier and several manufacturers with different ideal sequences. They describe efficient solution procedures for a variety of objectives. Dawande et al. (2006) study conflict and cooperation issues in a supply chain where a manufacturer makes products which are shipped to customers by a distributor. They evaluate conflict and describe efficient solution procedures.

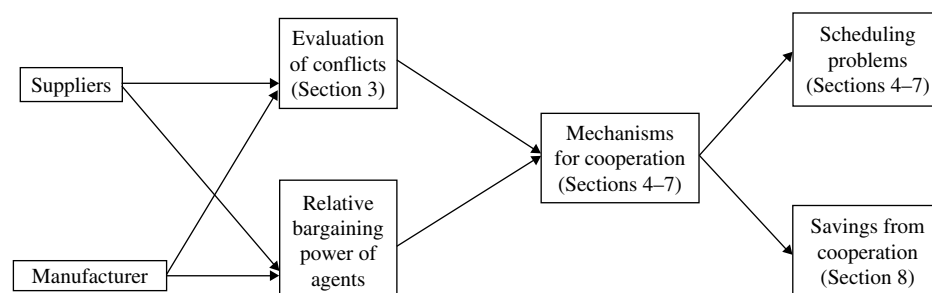
It is well known (Cachon 2003, Chen 2003) that cooperation between companies can improve efficiency. However, most supply chain cooperation models in the literature consider inventory control decisions. Hall and Potts (2003)

is apparently the only published paper that studies cooperation in supply chains with respect to scheduling decisions. Many electronics manufacturers, including Dell Computer, have recently adopted assemble-to-order production approaches (*The Economist* 2001) where the coordination of suppliers' production schedules with that of the manufacturer is critical.

In this paper, the issues of conflict and cooperation between the suppliers and the manufacturer arise from the requirement that all parts required for a job must be received before the final stage of manufacturing for that job to begin. Therefore, the performance of the manufacturer depends on the last supplier to deliver a part for each job, and thus on decisions made by all the suppliers. However, the suppliers are concerned about their own costs which are not, in general, optimized by the same scheduling decisions that optimize the manufacturer's cost. How these conflicts are resolved depends on the relative bargaining power of the suppliers and the manufacturer, and cooperation mechanisms that achieve this resolution need to be developed. We assume that all the suppliers act as a single agent when negotiating with the manufacturer. This is motivated by the common practice that sellers (or buyers) act together to negotiate collectively with buyers (or sellers) to increase their bargaining power (see, for example, the homepages of Alliance Suppliers Consortium (<http://asiconsort.net>) and Group Purchasing Advantage, Inc. (<http://www.grouppurchasingadvantage.com/>)). It is also necessary to evaluate the cost saving from cooperation between the decision makers to understand when cooperation is worthwhile. This in turn requires the solution of the various scheduling problems that these mechanisms define for solution by the suppliers, the manufacturer, and the overall system. Figure 1 provides an overview of the logical flow of this paper.

This paper is organized as follows. In §2, we specify our notation and assumptions. Section 3 contains an evaluation of the cost of conflicts between the solutions chosen by the suppliers and by the manufacturer. Because this cost is often significant, §§4 through 7 discuss how this conflict can be resolved under various assumptions about the relative bargaining power of the suppliers and the manufacturer, and identify cooperation mechanisms that are needed in each case. In §4, the suppliers dominate and

Figure 1. Logical flow of this paper.



the manufacturer offers them an incentive to obtain their cooperation and achieve a good solution. The reverse situation is considered in §5. In §6, the manufacturer dominates and moreover refuses to negotiate with the suppliers, who therefore need to adjust to the manufacturer's schedule. In §7, the suppliers and manufacturer have approximately equal bargaining power, and therefore they may cooperate to achieve a good solution. One motivation for studying all these different cases is that the resolution of conflict works differently in each case. In §8, we evaluate the cost saving that results from cooperation between the decision makers in our assembly system. Section 9 contains a discussion of extensions to allow bottleneck operations at the manufacturer and transportation times from the suppliers to the manufacturer. Finally, §10 contains a conclusion and some suggestions for future research. The missing proofs for the lemmas and theorems in this paper can be found in the electronic companion to this paper that is available as part of the online version and can be found at <http://or.journal.informs.org/>.

2. Notation and Assumptions

The assembly system studied in this paper consists of s suppliers, $S_1, \dots, S_s$, which supply parts to a manufacturer M . We assume that each supplier operates a dedicated production line for the manufacturer. This situation occurs commonly in practice, for example, when (1) the jobs from the manufacturer are customized and require special tools, (2) the manufacturer has more bargaining power than the suppliers, and (3) the supplier and manufacturer have a long-term relationship (*Electronic Business News* 2001). Each supplier may produce one or more parts for each job. Because our cost functions are nondecreasing in job completion time, there exists an optimal schedule in which the parts produced for a single job by the same supplier are scheduled consecutively.

The manufacturer receives a set of n jobs, $N = \{1, \dots, n\}$, from its customer(s) at the beginning of the planning horizon. Each job consists of several parts to be processed nonpreemptively at suppliers, $S_1, \dots, S_s$. The parts of a particular job can be processed concurrently by different suppliers. Because it has been shown in practice (Stein and Sweat 1998) and in the academic literature (Gavirneni et al. 1999) that sharing information vertically among the supply chain members helps the system operate more efficiently, we assume that the manufacturer passes all the known information about all the jobs to the suppliers simultaneously. Therefore, all parts become simultaneously available for processing at the suppliers. Let p_{ij} denote the total processing time of the parts processed by supplier S_i for job j for $i = 1, \dots, s$, $j = 1, \dots, n$. Each job j may have a due date d_j for $j = 1, \dots, n$. We assume that all parameters p_{ij} and d_j are nonnegative integers.

We define a *schedule* in any problem in this paper to consist of $s + 1$ subschedules, one for each of suppliers

$S_1, \dots, S_s$ and the manufacturer M . We define the following variables in a given schedule σ :

$C_{ij}(\sigma)$ = time at which the parts for job j are delivered from supplier S_i to the manufacturer;

$C_j(\sigma) = \max_{1 \leq i \leq s} \{C_{ij}(\sigma)\}$, the time at which the last part for job j is delivered to the manufacturer;

$L_j(\sigma) = C_j(\sigma) - d_j$, the lateness of job j .

We assume that the delivery time from a supplier to the manufacturer is negligible, relative to the processing times of the jobs at the supplier. The insights about conflict and cooperation issues which we discuss still hold without this assumption. However, the solvability of the implied scheduling problems may change in the more general case, as discussed in §9. Because we consider the manufacturer's operations as nonbottleneck, we assume without loss of generality that the processing time of each job at the manufacturer is zero. Therefore, $C_j(\sigma)$ is also the completion time of job j at the manufacturer. We use a shortest processing time (SPT) ordering of the parts or jobs, where they are scheduled in nondecreasing order of their processing times. We also use an earliest due date (EDD) ordering, where the jobs are scheduled in nondecreasing order of their due dates.

We consider various scheduling problems denoted by the classification scheme $\psi_1|\psi_2|\psi_3$ (Graham et al. 1979), where ψ_1 indicates the scheduling environment, ψ_2 describes the job characteristics or restrictive requirements, and ψ_3 defines the objective function to be minimized. Except where stated, we let $\psi_1 = As$, which denotes an assembly system with s suppliers and one manufacturer. We also refer to problems with $\psi_1 = 1$, which denotes a single-machine problem, or $\psi_1 = F2$, which denotes a two-machine flow-shop problem. Under ψ_2 , we may have *coseq*, which specifies that the suppliers use a common job sequence, t_{ij} , which denotes a transportation time for the parts of job j from supplier S_i to the manufacturer, or a constraint on cost.

To model conflict and cooperation issues in our assembly system, we consider four problems.

(1) *Supplier S_i 's problem.* An individual supplier S_i acts alone to minimize a performance measure F_i .

(2) *The suppliers' problem.* The suppliers act jointly to minimize a total performance measure $\sum_{i=1}^s F_i$.

(3) *The manufacturer's problem.* The manufacturer acts alone to minimize a performance measure G .

(4) *The system problem.* All the suppliers and the manufacturer act jointly to minimize their total cost $\alpha \sum_{i=1}^s F_i + (1 - \alpha)G$.

We consider the suppliers as a single agent, however the first definition helps in understanding the cost function of an individual supplier. In the system problem, α is used to convert the performance measures of the suppliers and the manufacturer into common cost units. For example, if one unit of $\sum_{i=1}^s F_i$ costs \$2 and one unit of G costs \$3, then $\alpha = 0.4$. Thus, αF_i represents the cost of supplier S_i

for $i = 1, \dots, s$, and $(1 - \alpha)G$ represents the cost of the manufacturer.

The models discussed here are applicable to an assembly system with any F_i and G cost functions. We now describe and motivate our specific choice of cost functions under ψ_3 . Because the suppliers do not deal directly with customers, their objective is internally focused; therefore, we use the minimization of work-in-process cost. We note that the total completion time of a set of jobs is often used in the scheduling literature (Lane and Sidney 1993, Ng et al. 2003) as a surrogate for the work-in-process inventory cost of the jobs. Furthermore, minimizing the total completion time of the jobs in a plant is equivalent to minimizing the average number of jobs in the plant and hence minimizing the congestion of the plant (Pinedo and Chao 1999, Hopp and Spearman 2000). For the manufacturer, however, customer service is important; therefore, we use total lead time and maximum lateness as cost measures. Total lead time measures average customer service, whereas maximum lateness measures worst case customer service. These scheduling cost functions are among the most commonly used in manufacturing practice. Other objective functions, such as the total tardiness or the total number of tardy jobs, could be considered in extensions of our work. Therefore, our discussion focuses on the following cost function F_i for supplier S_i for $i = 1, \dots, s$:

$\sum_{j \in N} C_{ij}(\sigma)$ = total completion time of the parts at supplier S_i in schedule σ , representing the work-in-process cost of the supplier,

and the following two cost functions G for the manufacturer:

$\sum_{j \in N} C_j(\sigma)$ = total completion time of the jobs at the manufacturer in schedule σ ;

$L_{\max}(\sigma) = \max\{L_j(\sigma) \mid j \in N\}$, maximum lateness of the jobs at the manufacturer in schedule σ .

When the schedule being used is clear from context, we write $C_{ij}(\sigma)$, $C_j(\sigma)$, and $L_{\max}(\sigma)$ as C_{ij} , C_j , and $L_{\max}$, respectively.

3. Conflicts Between Suppliers and Manufacturer

We study the extent to which an optimal schedule for the manufacturer's problem can be suboptimal for the suppliers' problem, and vice versa. Let ν and γ denote an optimal schedule for the suppliers' problem $As \parallel \sum_{i=1}^s F_i$ and for the manufacturer's problem $As \parallel G$, respectively. Also, let π denote an optimal schedule for the system problem, $As \parallel \alpha \sum_{i=1}^s F_i + (1 - \alpha)G$. Conflicts between the suppliers and the manufacturer are measured by the following relative errors between ν and γ : (i) $[\sum_{i=1}^s F_i(\gamma) - \sum_{i=1}^s F_i(\nu)] / \sum_{i=1}^s F_i(\nu)$, and (ii) $[G(\nu) - G(\gamma)] / G(\gamma)$, where (i) represents the relative error of the optimal schedule for the manufacturer's problem when used for the suppliers' problem, and (ii) represents the relative error of the

optimal schedule for the supplier's problem when used for the manufacturer's problem.

For the analysis in this section, we use the following notation:

$$\begin{aligned} p_j &= \max\{p_{ij} \mid i = 1, \dots, s\}; \\ N_g &= \{j \in N \mid p_j = g\}; \\ M_g &= \{j \in N \mid \sum_{i=1}^s p_{ij} = g\}; \\ N_{ig} &= \{j \in N \mid p_{ij} = g\}; \\ n_g &= |N_g| = \text{number of jobs } j \text{ with } p_j = g; \\ m_g &= |M_g| = \text{number of jobs } j \text{ with } \sum_{i=1}^s p_{ij} = g; \\ n_{ig} &= |N_{ig}| = \text{number of jobs } j \text{ with } p_{ij} = g; \\ r_g &= \min\{n_{ig} \mid i = 1, \dots, s\}; \\ m_{igh} &= \text{number of jobs } j \text{ with } p_{ij} = h \text{ and } p_j \leq g; \\ q_{gh} &= \max\{m_{igh} \mid i = 1, \dots, s\}. \end{aligned}$$

In §3.1, we assume that the cost functions of both the suppliers and the manufacturer are total completion time. Similar results are derived in §3.2 where the cost function of the suppliers is the total completion time and that of the manufacturer is maximum lateness. These results emphasize the need for cooperation between the suppliers and the manufacturer.

3.1. Completion Time vs. Completion Time

We first evaluate the relative error of the optimal schedule ν for the supplier's problem $As \parallel \sum \sum C_{ij}$ when it is used for the manufacturer's problem $As \parallel \sum C_j$, i.e.,

$$\frac{\sum_{j \in N} C_j(\nu) - \sum_{j \in N} C_j(\gamma)}{\sum_{j \in N} C_j(\gamma)}.$$

As a preliminary step, we analyze the solvability of these two problems. We define an individual shortest processing time (ISPT) schedule to be one obtained when supplier S_i schedules its parts in SPT order, i.e., such that $p_{i1} \leq \dots \leq p_{in}$ for $i = 1, \dots, s$.

THEOREM 1. *Problem $As \parallel \sum \sum C_{ij}$ is solved optimally by an ISPT order of the jobs, which can be found in $O(n \log n)$ time.*

THEOREM 2. *The recognition version of problem $As \parallel \sum C_j$ is unary NP-complete.*

In view of Theorem 2, we use schedule ρ generated by the following heuristic, instead of γ , to evaluate the performance of ν .

Heuristic SMPT. Schedule the jobs in nondecreasing order of p_j at all the suppliers, where $p_j = \max\{p_{ij} \mid i = 1, \dots, s\}$.

Clearly, $\sum_{j \in N} C_j(\gamma) \leq \sum_{j \in N} C_j(\rho)$. Therefore,

$$\begin{aligned} & \frac{\sum_{j \in N} C_j(\nu) - \sum_{j \in N} C_j(\gamma)}{\sum_{j \in N} C_j(\gamma)} \\ & \geq \frac{\sum_{j \in N} C_j(\nu) - \sum_{j \in N} C_j(\rho)}{\sum_{j \in N} C_j(\rho)}. \end{aligned} \quad (1)$$

The following result provides an asymptotic analysis for the relative error as $n \rightarrow \infty$.

THEOREM 3. Assume that the job processing times p_{ij} are integer valued and follow an independent and identical discrete uniform distribution over a finite interval $[l, u]$, where l and u are nonnegative integers. If the optimal schedule ν for the suppliers' problem is used for the manufacturer's problem $As \parallel \sum C_j$, then the relative error between the cost of ν and the optimal manufacturer's cost satisfies

$$\lim_{n \rightarrow \infty} \frac{\sum_{j \in N} C_j(\nu) - \sum_{j \in N} C_j(\gamma)}{\sum_{j \in N} C_j(\gamma)} \stackrel{a.s.}{\geq} \frac{\sum_{g=l}^u (g^2 - l^2 - g + l)[(g-l+1)^s - (g-l)^s]}{\sum_{g=l}^u [(g^2 - l^2 + g + l)(\frac{g-l+1}{u-l+1})^{s-1}][(g-l+1)^s - (g-l)^s]} - 1. \quad (2)$$

PROOF. From the laws of large numbers and the given assumption on the probability distribution of the p_{ij} values, we have, for $i = 1, \dots, s$, $g = l, \dots, u$, and $h = l, \dots, g$,

$$\lim_{n \rightarrow \infty} \frac{n_{ig}}{n} \stackrel{a.s.}{=} \Pr[p_{ij} = g] = \frac{1}{u-l+1}, \quad (3)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n_g}{n} &\stackrel{a.s.}{=} \Pr[p_j = g] = \Pr[p_j \leq g] - \Pr[p_j \leq g-1] \\ &= \left(\frac{g-l+1}{u-l+1}\right)^s - \left(\frac{g-l}{u-l+1}\right)^s, \quad \text{and} \end{aligned} \quad (4)$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{m_{igh}}{n} &\stackrel{a.s.}{=} \Pr[p_{ij} = h, p_j \leq g] \\ &= \Pr[p_j \leq g] \times \Pr[p_{ij} = h \mid p_j \leq g] \\ &= \left(\frac{g-l+1}{u-l+1}\right)^s \times \frac{1}{g-l+1} = \frac{(g-l+1)^{s-1}}{(u-l+1)^s}. \end{aligned} \quad (5)$$

It follows from (3) and (5) that

$$\lim_{n \rightarrow \infty} \frac{r_g}{n} \stackrel{a.s.}{=} \frac{1}{u-l+1} \quad \text{and} \quad (6)$$

$$\lim_{n \rightarrow \infty} \frac{q_{gh}}{n} \stackrel{a.s.}{=} \frac{(g-l+1)^{s-1}}{(u-l+1)^s}. \quad (7)$$

We provide a lower bound on the cost of ν in the manufacturer's problem. From Theorem 1, in schedule ν , the parts are processed at each supplier S_i in SPT order of their processing times p_{ij} . Thus, each supplier S_i processes the jobs from the same subset N_{ig} consecutively for $g = l, \dots, u$, and jobs from different subsets in the order $(N_{il}, N_{i,l+1}, \dots, N_{iu})$. Let E_{ig} denote the starting time of the first job from N_{ig} at supplier S_i . We have, for each $j \in N_g$,

$$\begin{aligned} C_j(\nu) &\geq \min\{E_{ig} \mid i = 1, \dots, s\} = \min\left\{\sum_{k=l}^{g-1} kn_{ik} \mid i = 1, \dots, s\right\} \\ &\geq \sum_{k=l}^{g-1} k \min\{n_{ik} \mid i = 1, \dots, s\} = \sum_{k=l}^{g-1} kr_k. \end{aligned}$$

This implies that

$$\sum_{j \in N} C_j(\nu) = \sum_{g=l}^u \sum_{j \in N_g} C_j(\nu) \geq \sum_{g=l}^u \left[n_g \sum_{k=l}^{g-1} kr_k \right]. \quad (8)$$

Then, from (4), (6), and (8), we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\sum_{j \in N} C_j(\nu)}{n^2} &\geq \lim_{n \rightarrow \infty} \sum_{g=l}^u \frac{n_g}{n} \left[\sum_{k=l}^{g-1} \frac{r_k}{n} k \right] \\ &= \sum_{g=l}^u \left(\lim_{n \rightarrow \infty} \frac{n_g}{n} \right) \left[\sum_{k=l}^{g-1} \left(\lim_{n \rightarrow \infty} \frac{r_k}{n} \right) k \right] \\ &\stackrel{a.s.}{=} \sum_{g=l}^u \left[\left(\frac{g-l+1}{u-l+1} \right)^s - \left(\frac{g-l}{u-l+1} \right)^s \right] \left[\sum_{k=l}^{g-1} \frac{k}{u-l+1} \right] \\ &= \frac{1}{2(u-l+1)^{s+1}} \sum_{g=l}^u [(g-l+1)^s - (g-l)^s] \\ &\quad \cdot [g^2 - l^2 - g + l]. \end{aligned} \quad (9)$$

Next, we derive an upper bound on the optimal manufacturer's cost, $\sum_{j \in N} C_j(\gamma)$, using schedule ρ from Heuristic SMPT and (1). In schedule ρ , the jobs from the same subset N_g for $g = 1, \dots, s$, are scheduled consecutively, and jobs from different subsets are scheduled in the order $(N_l, N_{l+1}, \dots, N_u)$. Let D_g denote the completion time of the last job from the subset N_g , and H_{ig} denote the total processing time of the jobs from $\bigcup_{k=l}^g N_k$ at supplier S_i . Thus, for $g = l, \dots, u$,

$$\begin{aligned} D_g &= \max\{H_{ig} \mid i = 1, \dots, s\} = \max\left\{\sum_{k=l}^g km_{igk} \mid i = 1, \dots, s\right\} \\ &\leq \sum_{k=l}^g k \max\{m_{igh} \mid i = 1, \dots, s\} = \sum_{k=l}^g kq_{gk}. \end{aligned} \quad (10)$$

Because $C_j(\rho) \leq D_g$ for $j \in N_g$, from (10), we have

$$\sum_{j \in N} C_j(\rho) = \sum_{g=l}^u \sum_{j \in N_g} C_j(\rho) \leq \sum_{g=l}^u n_g D_g \leq \sum_{g=l}^u n_g \left[\sum_{k=l}^g kq_{gk} \right]. \quad (11)$$

Then, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\sum_{j \in N} C_j(\gamma)}{n^2} &\leq \lim_{n \rightarrow \infty} \frac{\sum_{j \in N} C_j(\rho)}{n^2} \leq \lim_{n \rightarrow \infty} \sum_{g=l}^u \left[\frac{n_g}{n} \sum_{k=l}^g \frac{q_{gk}}{n} k \right] \quad \text{from (11)} \\ &= \sum_{g=l}^u \left[\left(\lim_{n \rightarrow \infty} \frac{n_g}{n} \right) \sum_{k=l}^g \left(\lim_{n \rightarrow \infty} \frac{q_{gk}}{n} \right) k \right] \\ &\stackrel{a.s.}{=} \sum_{g=l}^u \left[\frac{(g-l+1)^s - (g-l)^s}{(u-l+1)^s} \right] \sum_{k=l}^g \left[\frac{(g-l+1)^{s-1}}{(u-l+1)^s} k \right] \\ &\quad \text{from (4) and (7)} \end{aligned}$$

$$= \frac{1}{2(u-l+1)^{s+1}} \sum_{g=l}^u [(g-l+1)^s - (g-l)^s] \cdot [g^2 - l^2 + g + l] \left[\frac{(g-l+1)^{s-1}}{(u-l+1)^{s-1}} \right]. \quad (12)$$

The main result, (2), then follows from (9) and (12). $\square$

Theorem 3 provides an asymptotic lower bound on the relative error between schedules ν and γ for the manufacturer's problem, as $n \rightarrow \infty$. For finite values of n , we compute the lower bound in (1) using random test problems which are generated as follows: (i) number of suppliers $s \in \{2, 4, 8, 16\}$; (ii) number of jobs $n \in \{50, 100, 200, 400\}$; and (iii) job processing times p_{ij} are independently drawn from a uniform integer distribution over the interval $[l, u]$, where $l \in \{10, 50\}$ and $u \in \{100, 500, 1,000\}$. For a small number of randomly generated test problems, we have $\sum_{j \in N} C_j(\nu) < \sum_{j \in N} C_j(\rho)$. For these problems, we record the relative error as zero because the actual relative error between ν and γ is always nonnegative. For each (s, l, u, n) combination, the mean lower bound over 10 random instances is reported in Table 1. We also compute the lower bound in (2) for each (s, l, u) combination and report the results in the columns with $n = \infty$ in Table 1. Because the relative error is at least 19.6% for all the (s, l, u, n) combinations considered, we conclude that there is substantial

Table 1. Lower bounds on relative error in the manufacturer's problem ($\sum C_j$).

s	l	u	Lower bound on relative error				
			$n = 50$ (%)	$n = 100$ (%)	$n = 200$ (%)	$n = 400$ (%)	$n = \infty$ (%)
2	10	100	21.94	23.29	24.42	25.13	22.66
		500	19.91	21.29	22.68	23.52	24.58
		1,000	19.63	20.99	22.42	23.30	24.79
4	10	100	39.34	43.98	45.70	47.32	45.73
		500	36.45	41.62	43.38	45.09	49.21
		1,000	35.99	41.21	43.04	44.80	49.61
8	10	100	54.34	57.11	61.53	65.08	61.18
		500	50.94	53.89	58.80	62.99	68.30
		1,000	50.48	53.46	58.47	62.70	69.15
16	10	100	62.63	67.95	72.80	75.24	66.35
		500	59.42	64.71	70.49	73.24	79.93
		1,000	59.00	64.43	70.25	72.96	81.63
2	50	100	28.55	29.58	29.98	30.28	24.89
		500	21.93	23.31	24.48	25.22	25.97
		1,000	20.74	22.08	23.40	24.21	25.55
4	50	100	50.03	52.59	53.85	54.53	47.29
		500	39.49	44.12	45.83	47.33	50.71
		1,000	37.65	42.60	44.36	45.99	50.43
8	50	100	66.33	68.22	70.56	72.38	58.84
		500	54.20	57.03	61.39	65.15	69.33
		1,000	52.23	55.12	59.87	63.85	69.74
16	50	100	75.54	78.42	81.56	82.44	57.97
		500	62.70	67.66	72.95	75.28	80.38
		1,000	60.79	65.87	71.49	74.04	81.95

conflict between the two schedules. The results in Table 1 indicate that the relative errors increase significantly with the number of suppliers s . This can be explained as follows. As s increases, C_j increases because it is the maximum of C_{ij} over $i = 1, \dots, s$. Also, the increase of C_j under schedule ν is likely to be larger than that under schedule ρ because schedule ρ uses the same job sequence at all the suppliers, whereas schedule ν does not. The relative errors also increase slightly with n for $50 \leq n \leq 400$.

We next evaluate the relative error of the optimal schedule γ for the manufacturer's problem $As \parallel \sum C_j$ when it is used for the suppliers' problem $As \parallel \sum \sum C_{ij}$, i.e.,

$$\frac{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\gamma) - \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}.$$

As a preliminary step, we analyze the solvability of these two problems. In view of Theorems 1 and 2, we use schedule ξ generated by the following shortest total processing time rule to evaluate the performance of γ .

Heuristic STPT. Schedule the jobs in nondecreasing order of $\sum_{i=1}^s p_{ij}$ at each supplier.

We need the following two preliminary results to provide some insight.

THEOREM 4. *Heuristic STPT finds an optimal schedule for problem $As | coseq | \sum \sum C_{ij}$ in $O(n \log n)$ time.*

LEMMA 1. *For any manufacturer's problem with a regular objective function of $(C_1, \dots, C_n)$ denoted by $As | \cdot | \max_{1 \leq j \leq n} \{f_j(C_j)\}$ or $As | \cdot | \sum f_j(C_j)$, where $f_j(\cdot)$ is a non-decreasing function, and without processing times at the manufacturer or transportation times, there exists an optimal schedule with identical job sequences at all suppliers.*

From Theorem 4, schedule ξ minimizes $\sum \sum C_{ij}$ over all schedules that require identical job sequences at the suppliers. From Lemma 1, we can assume without loss of generality that schedule γ has identical job sequences at the suppliers. Therefore, we have

$$\sum_{i=1}^s \sum_{j \in N} C_{ij}(\gamma) \geq \sum_{i=1}^s \sum_{j \in N} C_{ij}(\xi) \quad (13)$$

$$\Rightarrow \frac{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\gamma) - \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)} \geq \frac{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\xi) - \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}. \quad (14)$$

The following result provides an asymptotic analysis for the relative error as $n \rightarrow \infty$.

THEOREM 5. *Assume that the job processing times p_{ij} are integer valued and follow an independent and identical discrete uniform distribution over a finite interval $[l, u]$, where l and u are nonnegative integers. If the optimal schedule γ for the manufacturer's problem is used for the suppliers'*

problem $As\|\sum\sum C_{ij}$, then the relative gap between the cost of γ and the optimal manufacturer's cost satisfies

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\gamma) - \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)} \geq \frac{\sum_{g=sl}^{su} [r_{sg} \sum_{h=sl}^{g-1} h r_{sh}]}{s[\sum_{g=l}^u (g^2 - l^2 + g + l)]/[2(u-l+1)^2]} - 1, \quad (15)$$

where r_{sk} is the probability that the total processing time $\sum_{i=1}^s p_{ij}$ of any job is equal to k for $sl \leq k \leq su$.

Theorem 5 provides an asymptotic lower bound on the relative gap between schedules γ and ν for the suppliers' problem, as $n \rightarrow \infty$. The probability parameters r_{sk} can be calculated recursively for given l and u , as follows. First, when $s = 1$, we have $r_{1k} = 1/(u-l+1)$ for $l \leq k \leq u$. Suppose that we have already calculated r_{sk} for $s = 1, \dots, i$ for some i . Then, the values of $r_{i+1,k}$ can be found from the following formula: $r_{i+1,k} = \sum_{j=l}^{\min(u,k)} r_{i,k-j}/(u-l+1)$ for $(i+1)l \leq k \leq (i+1)u$. For finite values of n , we compute the lower bound in (14) using random test problems which are generated similarly to those in Table 1. The results are reported in Table 2. We also compute the lower bound in (15) for each (s, l, u) combination and report the results in the columns with $n = \infty$ in Table 2. Because the relative error is at least 10% for most (s, l, u, n) combinations

Table 2. Lower bounds on relative error in the suppliers' problem ($\sum\sum C_{ij}$).

s	l	u	Lower bound on relative error				
			$n = 50$ (%)	$n = 100$ (%)	$n = 200$ (%)	$n = 400$ (%)	$n = \infty$ (%)
2	10	100	10.60	11.13	11.26	11.23	8.73
		500	13.16	13.80	13.99	13.97	13.64
		1,000	13.53	14.19	14.39	14.37	14.31
4	10	100	17.29	17.94	18.74	18.78	16.82
		500	21.42	22.28	23.31	23.35	23.50
		1,000	22.02	22.91	23.97	24.01	24.44
8	10	100	23.56	24.48	24.47	24.60	22.46
		500	29.19	30.41	30.41	30.59	30.38
		1,000	30.01	31.27	31.27	31.46	31.51
16	10	100	27.27	27.95	28.47	28.46	26.43
		500	33.81	34.70	35.38	35.38	35.21
		1,000	34.76	35.68	36.38	36.39	36.47
2	50	100	3.56	3.72	3.74	3.72	0.06
		500	10.50	11.00	11.14	11.11	10.74
		1,000	12.05	12.64	12.81	12.78	12.70
4	50	100	5.82	5.98	6.22	6.24	3.07
		500	17.12	17.76	18.55	18.59	18.61
		1,000	19.64	20.41	21.33	21.37	21.73
8	50	100	7.90	8.16	8.15	8.18	5.18
		500	23.32	24.24	24.23	24.36	24.09
		1,000	26.76	27.85	27.85	28.00	28.01
16	50	100	9.13	9.33	9.48	9.46	6.67
		500	27.00	27.68	28.19	28.18	27.94
		1,000	30.99	31.79	32.40	32.39	32.43

considered, we conclude that there is substantial conflict between the two schedules. Because in (14) schedules γ and ξ use a common sequence for all the suppliers, whereas schedule ν does not, the relative errors increase with s .

3.2. Completion Time vs. Maximum Lateness

We first evaluate the performance of the optimal schedule ν for the supplier's problem $As\|\sum\sum C_{ij}$ when it is used for the manufacturer's problem $As\|L_{\max}$, i.e.,

$$\frac{L_{\max}(\nu) - L_{\max}(\gamma)}{L_{\max}(\gamma)}.$$

As a preliminary step, we briefly discuss the solvability of these two problems. Problem $As\|\sum\sum C_{ij}$ is studied in Theorem 1. For problem $As\|L_{\max}$, the following result generalizes that of Jackson (1955) for the single-machine scheduling problem $1\|L_{\max}$.

THEOREM 6. An optimal schedule for problem $As\|L_{\max}$ is provided by an earliest due date (EDD) ordering of the jobs, which can be found in $O(n \log n)$ time.

The following result provides an asymptotic analysis for the relative error as $n \rightarrow \infty$.

THEOREM 7. Assume that

(i) the job processing times p_{ij} are integer valued and follow an independent and identical discrete uniform distribution over a finite interval $[l, u]$, where l and u are nonnegative integers,

(ii) the job due dates d_j follow an independent and identical uniform distribution over the interval $[0, n\bar{p}]$, where $\bar{p} = (l+u)/2$ is the average processing time of a job at a supplier, and

(iii) the p_{ij} and d_j values are independent.

If the optimal schedule ν for the suppliers' problem is used for the manufacturer's problem $As\|L_{\max}$, then the relative error between the maximum lateness of ν and the optimal manufacturer's maximum lateness satisfies

$$\lim_{n \rightarrow \infty} \frac{L_{\max}(\nu) - L_{\max}(\gamma)}{L_{\max}(\gamma)} \stackrel{a.s.}{=} \infty. \quad (16)$$

Theorem 7 means that as $n \rightarrow \infty$, the relative error between schedule ν and γ for the manufacturer's problem becomes, almost surely, arbitrarily large. For finite values of n , we compute the relative error using random test problems where the parameters s , n , l , u , and p_{ij} are generated similarly to those in Tables 1 and 2, and where the parameter d_j is independently drawn from a discrete uniform distribution over the interval $[0, n(l+u)/2]$. For each (s, l, u, n) combination, the mean relative error over 10 random instances is reported in Table 3. As can be seen there, large relative errors occur over all (s, l, u, n) combinations considered, indicating huge conflict between the two schedules. These errors are much larger than those observed in Tables 1 and 2 because lower bounds on the value of $L_{\max}$

Table 3. Relative errors in the manufacturer's problem ($L_{\max}$).

s	l	u	Relative error				
			$n = 50$ (%)	$n = 100$ (%)	$n = 200$ (%)	$n = 400$ (%)	$n = \infty$ (%)
2	10	100	764	1,006	1,465	2,059	∞
		500	636	875	1,288	1,765	∞
		1,000	621	864	1,270	1,730	∞
4	10	100	627	1,084	1,258	2,227	∞
		500	552	936	1,062	1,769	∞
		1,000	543	917	1,042	1,722	∞
8	10	100	603	873	1,337	2,286	∞
		500	511	739	1,078	1,814	∞
		1,000	504	724	1,050	1,771	∞
16	10	100	436	786	1,185	1,992	∞
		500	372	677	952	1,565	∞
		1,000	366	660	930	1,515	∞
2	50	100	1,305	1,419	2,066	2,962	∞
		500	722	956	1,397	1,730	∞
		1,000	662	905	1,322	1,803	∞
4	50	100	991	1,542	2,035	3,633	∞
		500	611	999	1,189	1,978	∞
		1,000	572	958	1,103	1,825	∞
8	50	100	1,013	1,398	2,148	3,964	∞
		500	582	830	1,223	2,035	∞
		1,000	538	770	1,118	1,875	∞
16	50	100	669	1,317	2,361	3,739	∞
		500	421	763	1,087	1,789	∞
		1,000	391	703	992	1,621	∞

may be small. The relative errors increase strongly with n because as the number of jobs increases, the error caused by a single job with large processing times but a small due date being at the end of schedule ν increases.

We next evaluate the relative error of the optimal schedule γ for the manufacturer's problem $As\|L_{\max}$ when it is used for the suppliers' problem $As\|\sum \sum C_{ij}$, i.e.,

$$\frac{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\gamma) - \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}.$$

The following result provides an asymptotic analysis of the relative error as $n \rightarrow \infty$.

THEOREM 8. Assume that

(i) the job processing times p_{ij} are integer valued and follow an independent and identical discrete uniform distribution over a finite interval $[l, u]$, where l and u are nonnegative integers,

(ii) the job due dates d_j follow an arbitrary independent and identical distribution, and

(iii) the p_{ij} and d_j values are independent.

If the optimal schedule γ for the manufacturer's problem is used for the suppliers' problem $As\|\sum \sum C_{ij}$, then the

relative error between the cost of γ and that of ν satisfies

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\gamma) - \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)}{\sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu)} \geq \frac{a.s.}{2} \frac{(u+l)(u-l+1)^2}{\sum_{g=l}^u (g^2 - l^2 + g + l)} - 1. \quad (17)$$

Theorem 8 provides an asymptotic lower bound on the relative error between schedules γ and ν for the suppliers' problem. For finite values of n , we compute the relative error using random test problems which are generated similarly to those in Table 3. For each (s, l, u, n) combination, the mean relative error over 10 random instances is reported in Table 4. We also compute the lower bound in (17) for each (s, l, u) combination and report the results in the columns with $n = \infty$ in Table 4. Because the relative error is at least 10.3% for all (s, l, u, n) combinations considered, we conclude that there is substantial conflict between the schedules. The explanation of the results follows that for Table 2.

4. Suppliers Dominate, Manufacturer Negotiates

Motivated by the evidence of significant cost of conflict found in §3, we investigate in §§4 through 7 how different bargaining power scenarios lead to various resolutions

Table 4. Relative errors in the suppliers' problem ($\sum \sum C_{ij}$).

s	l	u	Relative error				
			$n = 50$ (%)	$n = 100$ (%)	$n = 200$ (%)	$n = 400$ (%)	$n = \infty$ (%)
2	10	100	34.78	36.19	36.82	37.17	36.01
		500	43.11	44.96	45.83	46.20	46.82
		1,000	44.31	46.25	47.15	47.52	48.38
4	10	100	36.50	37.23	37.44	37.64	36.01
		500	45.26	46.28	46.50	46.77	46.82
		1,000	46.53	47.60	47.82	48.10	48.38
8	10	100	35.99	37.03	37.66	37.65	36.01
		500	44.65	45.96	46.78	46.79	46.82
		1,000	45.90	47.26	48.10	48.12	48.38
16	10	100	36.06	37.19	37.59	37.80	36.01
		500	44.68	46.15	46.70	47.01	46.82
		1,000	45.93	47.45	48.03	48.35	48.38
2	50	100	11.64	12.02	12.19	12.35	10.34
		500	34.40	35.80	36.44	36.79	37.20
		1,000	39.50	41.16	41.93	42.30	43.03
4	50	100	12.20	12.37	12.47	12.53	10.34
		500	36.11	36.84	37.06	37.26	37.20
		1,000	41.47	42.36	42.59	42.83	43.03
8	50	100	12.02	12.36	12.55	12.52	10.34
		500	35.62	36.66	37.28	37.27	37.20
		1,000	40.91	42.11	42.84	42.84	43.03
16	50	100	12.09	12.42	12.52	12.55	10.34
		500	35.70	36.82	37.22	37.42	37.20
		1,000	40.96	42.28	42.77	43.03	43.03

of this conflict. Figure 1 shows how this discussion fits within the paper. In this section, we consider a situation where the suppliers have dominant bargaining power in the supply chain and choose their optimal schedule, ν . Here, the manufacturer may wish to compensate the suppliers for using some alternative schedule σ which is cheaper for the manufacturer than ν . The bargaining theory of Nash (1950, 1953) suggests that this incentive has to be enough to persuade the suppliers to select schedule σ (to be determined) instead of ν , while the resulting cost to the manufacturer has to be no more than that of ν . Recall that αF_i denotes the cost of supplier S_i , and $(1 - \alpha)G$ denotes the cost of the manufacturer. Therefore, the incentive u_i provided to each supplier S_i must satisfy

$$\alpha F_i(\sigma) - u_i \leq \alpha F_i(\nu) \quad \text{for } i = 1, \dots, s, \quad \text{and} \quad (18)$$

$$(1 - \alpha)G(\sigma) + \sum_{i=1}^s u_i \leq (1 - \alpha)G(\nu). \quad (19)$$

Also, the manufacturer chooses σ to minimize its total cost $(1 - \alpha)G(\sigma) + \sum_{i=1}^s u_i$. Thus, the manufacturer is faced with the following optimization problem in determining σ and $u_1, \dots, u_s$:

$$\begin{aligned} \min_{\sigma, u_1, \dots, u_s} \quad & (1 - \alpha)G(\sigma) + \sum_{i=1}^s u_i \\ \text{s.t.} \quad & (18), (19), \text{ and } u_i \geq 0 \text{ for } i = 1, \dots, s. \end{aligned} \quad (20)$$

THEOREM 9. *The solution $(\sigma, u_1, \dots, u_s)$ defined by $\sigma = \pi$ and $u_i = \alpha[F_i(\pi) - F_i(\nu)]$ for $i = 1, \dots, s$, where π is the schedule that minimizes $\alpha \sum_{i=1}^s F_i(\sigma) + (1 - \alpha)G(\sigma)$, is optimal for problem (20).*

PROOF. By definition of π , we have $\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \leq \alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)G(\nu)$. Thus, the solution $(\sigma, u_1, \dots, u_s)$ satisfies (19). Because by construction it also satisfies (18), it follows that this solution is feasible for problem (20). From (18), given any feasible solution $(\sigma, u_1, \dots, u_s)$ to problem (20), its cost to the manufacturer, denoted by $Z(\sigma, u_1, \dots, u_s)$, satisfies

$$\begin{aligned} Z(\sigma, u_1, \dots, u_s) \\ \geq \alpha \sum_{i=1}^s F_i(\sigma) + (1 - \alpha)G(\sigma) - \alpha \sum_{i=1}^s F_i(\nu). \end{aligned} \quad (21)$$

Because $\alpha \sum_{i=1}^s F_i(\sigma) + (1 - \alpha)G(\sigma)$ is minimized when $\sigma = \pi$, the right-hand side of (21) is minimized when $\sigma = \pi$. Therefore, $\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) - \alpha \sum_{i=1}^s F_i(\nu)$ is a lower bound on optimal cost in problem (20). However, the cost of the solution given in the theorem equals this lower bound. Therefore, this solution is optimal for problem (20). $\square$

Theorem 9 implies that (i) problem (20) is feasible and hence the proposed cooperation mechanism can always be used; and (ii) the optimal strategy of the manufacturer is

to provide an incentive of $u_i = \alpha[F_i(\pi) - F_i(\nu)]$ to each supplier S_i for $i = 1, \dots, s$. Consequently, the suppliers agree to use schedule π . It follows that the manufacturer's cost is equal to the optimal cost in problem (20), i.e., $\alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] + (1 - \alpha)G(\pi)$. In the absence of cooperation, schedule ν is used and the manufacturer's cost is $(1 - \alpha)G(\nu)$. Thus, the net saving to the manufacturer from obtaining the suppliers' cooperation is

$$\begin{aligned} (1 - \alpha)G(\nu) - \alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] - (1 - \alpha)G(\pi) \\ = \left[\alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)G(\nu) \right] \\ - \left[\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \right], \end{aligned} \quad (22)$$

which is also the difference between the total system cost under schedule ν and under schedule π . Thus, if ν is not optimal for the system problem, then the net saving to the manufacturer in (22) is positive, and hence the manufacturer should encourage the suppliers to use π by providing the incentive u_i given in Theorem 9 to each supplier S_i for $i = 1, \dots, s$. However, if ν is optimal for the system problem, then no incentive provided to the suppliers creates a net saving to the manufacturer. In this case, the suppliers use schedule ν . In either case, the suppliers' total cost is no greater than the optimal cost given by schedule ν . Because the suppliers use a schedule which is optimal for the system problem (i.e., either π , or ν if ν is optimal for the system problem), our proposed cooperation mechanism is equivalent to centralized decision making for the system as a whole.

Theorem 9 shows that, whether $\pi = \nu$ or not, the optimal system schedule π is always achieved. Therefore, we need to operationalize this result by studying the solvability of the two system problems defined by our cooperation mechanism. In doing so, we note that α is fixed, therefore the system problem does not directly generalize the related suppliers' or manufacturer's problem.

First, we consider problem $As \parallel \alpha \sum \sum C_{ij} + (1 - \alpha) \sum C_j$.

THEOREM 10. *The recognition version of problem $As \parallel \alpha \sum \sum C_{ij} + (1 - \alpha) \sum C_j$, where $0 < \alpha < 1$ is a fixed parameter, is unary NP-complete.*

Theorem 10 motivates the use of ISPT as a heuristic for problem $As \parallel \alpha \sum \sum C_{ij} + (1 - \alpha) \sum C_j$.

THEOREM 11. *For problem $As \parallel \alpha \sum \sum C_{ij} + (1 - \alpha) \sum C_j$, let the value of the solution delivered by ISPT be F^H , and the optimal objective value be F^* . Then, $F^H \leq [s/(\alpha s - \alpha + 1)]F^*$.*

It follows from Theorem 11 that Heuristic ISPT is optimal for problem $As \parallel \sum \sum C_{ij}$, and also performs well for problem $As \parallel \alpha \sum \sum C_{ij} + (1 - \alpha) \sum C_j$ when α is close to

one. Alternatively, when α is close to zero, the problem is similar to $As\|\sum C_j$, for which we discuss a heuristic in §6.

Second, we consider problem $As\|\alpha\sum\sum C_{ij} + (1-\alpha) \cdot L_{\max}$. Hoogeveen and van de Velde (1995) describe an $O(n^3 \log n)$ time algorithm for finding the Pareto optimal solutions for problem $1\|F(\sum C_j, L_{\max})$, where F denotes an arbitrary linear additive objective function, and C_j denotes the completion time of job j on the single machine. We propose the following algorithm.

Algorithm SCL

Step 1. For each supplier, apply the algorithm of Hoogeveen and van de Velde (1995) to find all the Pareto optimal $(\sum C_j, L_{\max})$ pairs.

Step 2. Sort the Pareto-optimal solutions for each supplier $i = 1, \dots, s$ by nondecreasing order of $L_{\max}$ values. Let $(l_{i1}, \dots, l_{ik_i})$ and $(\pi_{i1}, \dots, \pi_{ik_i})$ denote the resulting list of $L_{\max}$ values and the associated schedules of supplier S_i , respectively, where k_i is the number of Pareto-optimal solutions for supplier S_i . Merge the lists $(l_{i1}, \dots, l_{ik_i})$ for $i = 1, \dots, s$ to create a single list $(l_{[1]}, \dots, l_{[q]})$, where $l_{[1]} \leq \dots \leq l_{[q]}$, and q is the number of different values of l_{ih} for $i = 1, \dots, s$ and $h = 1, \dots, q$.

Step 3. For each $l_{[h]}$, for $h = 1, \dots, q$, do the following:

For each $i = 1, \dots, s$, find index u_i from the list $(l_{i1}, \dots, l_{ik_i})$ such that $l_{u_i} \leq l_{[h]}$ and l_{u_i} is closest to $l_{[h]}$ in this list.

Compute the total cost $\alpha \sum_{i=1}^s \sum_{j \in N} C_{ij}(\pi_{iu_i}) + (1-\alpha)l_{[h]}$.

Select a solution with minimum total cost.

THEOREM 12. *Algorithm SCL solves problem $As\|\alpha\sum\sum C_{ij} + (1-\alpha)L_{\max}$ in $O(sn^3 \log n)$ time.*

5. Manufacturer Dominates, Suppliers Negotiate

In the context of Figure 1, we consider a situation where the manufacturer has dominant bargaining power and chooses its optimal schedule γ . Here, the suppliers may wish the manufacturer to choose some other schedule which is cheaper than γ for the suppliers. To resolve this conflict, we propose the following cooperation mechanism.

The suppliers together offer an incentive to the manufacturer to use a schedule that is better than γ . Let v and σ denote the incentive provided by the suppliers, and the schedule which the suppliers want the manufacturer to use, respectively. Similarly to the derivation of (20), the suppliers are faced with the following optimization problem:

$$\min_{\sigma, v} \alpha \sum_{i=1}^s F_i(\sigma) + v \quad (23)$$

$$\text{s.t. } (1-\alpha)G(\sigma) - v \leq (1-\alpha)G(\gamma), \quad (24)$$

$$\alpha \sum_{i=1}^s F_i(\sigma) + v \leq \alpha \sum_{i=1}^s F_i(\gamma), \quad (25)$$

$$v \geq 0. \quad (26)$$

THEOREM 13. *The solution (σ, v) defined by $\sigma = \pi$ and $v = (1-\alpha)[G(\pi) - G(\gamma)]$ is optimal for problem (23)–(26), where π is the schedule that minimizes $\alpha \sum_{i=1}^s F_i(\sigma) + (1-\alpha)G(\sigma)$.*

PROOF. The proof here is similar to that of Theorem 9. $\square$

Theorem 13 implies that (i) problem (23)–(26) is feasible, therefore the proposed cooperation mechanism can always be used if the suppliers are willing to negotiate collectively with the manufacturer; and (ii) the optimal strategy of the suppliers is to provide collectively an incentive of $v = (1-\alpha)[G(\pi) - G(\gamma)]$ to the manufacturer. Consequently, the manufacturer agrees to use schedule π , and the net saving to the suppliers from the manufacturer's cooperation is given by

$$\begin{aligned} \Delta &= \alpha \sum_{i=1}^s F_i(\gamma) - \left\{ \alpha \sum_{i=1}^s F_i(\pi) + (1-\alpha)[G(\pi) - G(\gamma)] \right\} \\ &= \left[\alpha \sum_{i=1}^s F_i(\gamma) + (1-\alpha)G(\gamma) \right] \\ &\quad - \left[\alpha \sum_{i=1}^s F_i(\pi) + (1-\alpha)G(\pi) \right] \geq 0. \end{aligned} \quad (27)$$

The first and second terms in (27) are the suppliers' total cost without and with the manufacturer's cooperation, respectively. If γ is not optimal for the system problem, then the net saving to the suppliers in (27) is positive, and hence the suppliers as a whole benefit from the manufacturer's cooperation. Here, the suppliers work together to provide the incentive v in Theorem 13 to the manufacturer, so that the manufacturer chooses schedule π . Theorem 13 shows that the suppliers prefer schedule π to γ . If γ is optimal for the system problem, then the suppliers do not bargain with the manufacturer because they as a whole do not benefit from the manufacturer's cooperation.

6. Manufacturer Dominates, Suppliers Adjust

Again in the context of Figure 1, we recognize the reality that, in some practical supply chains, the manufacturer may not allow its suppliers to bargain. This occurs when a manufacturer is much larger and/or financially stronger than all of its suppliers. Kraemer and Dedrick (2002) discuss the example of Dell Computer. Here, because the suppliers have no bargaining power, they need to meet any requirements imposed by the manufacturer without compensation. We consider situations where the manufacturer's dominance allows it to impose two natural requirements on the suppliers.

The first requirement arises from the manufacturer's choice of production sequence, γ . This requirement can be interpreted in two alternative ways: either that the suppliers face due dates implied by γ for all parts; or that some

delays in the manufacturer's schedule are permitted, subject to a predetermined limit on the resulting cost increase to the manufacturer. In either case, this requirement can be implemented by the suppliers individually, without cooperation between them.

The second requirement is that the suppliers all send the jobs to the manufacturer in some common sequence, so that the manufacturer's average lead time is minimized (see Lemma 1). One possible cooperation strategy for the suppliers is as follows. They find a schedule with the same sequence at each supplier which minimizes the total cost of the suppliers over all such sequences. We denote the problem of finding such a schedule by $As|coseq|\sum_{i=1}^s F_i$. We consider problem $As|coseq|\sum_{i=1}^s F_i$ with various objective functions. Let μ denote an optimal schedule to problem $As|coseq|\sum_{i=1}^s F_i$, which is chosen by the suppliers. Because schedule μ is by definition no better than schedule ν for supplier S_i , the cost of each supplier S_i increases by $\alpha[F_i(\mu) - F_i(\nu)] \geq 0$ when using μ instead of ν . Suppliers with a lower than average cost increase may need to compensate other suppliers.

To understand the scheduling problems faced by the suppliers as a result of the above requirements imposed by the manufacturer, we now discuss the solvability of the two manufacturer's problems, $As||\sum C_j$ and $As||L_{\max}$. Recall that Theorem 2 establishes the intractability of problem $As||\sum C_j$. Therefore, we now describe a heuristic, Batch, for this problem. We demonstrate that, under the following mild Assumption A about the job processing times, Batch provides solutions that are asymptotically optimal in the number of jobs.

ASSUMPTION A. *The processing times p_{ij} take only integer values and follow an independent and identical discrete distribution $\Phi(\cdot)$ over a finite interval $[L_p, U_p]$, where L_p and U_p are nonnegative integers.*

The intuition behind Heuristic Batch is to schedule most of the jobs in nondecreasing order of $\sum_{i=1}^s p_{ij}$, and use special sequences within jobs that have identical value of $\sum_{i=1}^s p_{ij}$. The heuristic first divides the jobs into different subsets based on their processing times. We refer to the vector of data $(p_{1j}, \dots, p_{sj})$ associated with a job j as the *parameter vector* of this job. We divide the jobs in N into different *classes* such that all the jobs in a class have the same parameter vector, i.e., they are identical. Let $N_1, \dots, N_K$ denote those classes of jobs, where $N = \bigcup_{k=1}^K N_k$. Note that K is finite even when n goes to infinity. The following $O(sK^2)$ time subroutine partitions the set N into classes $N_1, \dots, N_K$.

Subroutine Class

Step 0. Define the first class by $N_1 = \{1\}$. Let $K = 1$.

Step 1. For $j = 2, \dots, n$, compare the parameter vector of job j with that of each of the K classes already created, $N_1, \dots, N_K$. If job j is identical to the jobs in N_k for some k ,

$1 \leq k \leq K$, then let $N_k = N_k \cup \{j\}$. Otherwise, create a new class, $N_{K+1} = \{j\}$, and let $K = K + 1$.

Given a parameter vector $(x_1, x_2, \dots, x_s)$, we define $s - 1$ *shifted vectors* (Kaminsky and Simchi-Levi 1998) as follows:

$$(x_2, x_3, \dots, x_s, x_1), (x_3, x_4, \dots, x_s, x_1, x_2), \dots, (x_s, x_1, \dots, x_{s-1}).$$

We use the following $O(sK^2)$ time subroutine to combine the job classes $N_1, \dots, N_K$ into $H \leq K$ groups, $G_1, \dots, G_H$, such that the parameter vector of each class in a group is a shifted vector of each of the other classes in this group. It follows that there are at most s classes in a group. We denote $G_h = \{Q_{1h}, \dots, Q_{sh}\}$ where, for each $i = 1, \dots, s$, Q_{ih} is either empty or $Q_{ih} = N_k$ for some k with $1 \leq k \leq K$. Note that H is finite even when $n \rightarrow \infty$.

Subroutine Group

Step 0. Define the first group by $G_1 = \{N_1\}$. Let $H = 1$.

Step 1. For $k = 2, \dots, K$, compare the parameter vector of class N_k with that of each class in each group already created, $G_1, \dots, G_H$. If the parameter vector of class N_k is a shifted vector of that of a class in some group G_h for $1 \leq h \leq H$, then let $G_h = G_h \cup \{N_k\}$. Otherwise, create a new group, $G_{H+1} = \{N_k\}$, and let $H = H + 1$.

By construction, each job j in each group G_h (i.e., $j \in \bigcup_{i=1}^s Q_{ih}$) has identical total processing time $P_h = \sum_{i=1}^s p_{ij}$. We assume for the remainder of this section that groups $G_1, \dots, G_H$ are indexed such that $P_1 \leq \dots \leq P_H$. Let n_{ih} denote the number of jobs in class Q_{ih} , and define $n_h = \min\{n_{ih} \mid 1 \leq i \leq s\}$. Also, let $g_h = \sum_{i=1}^s n_{ih}$ denote the total number of jobs in group G_h , and define $\delta_h = \sum_{i=1}^s (n_{ih} - n_h)$.

Heuristic Batch

Step 0. Divide the jobs of N into the H groups $G_1, \dots, G_H$ as described above, by calling Subroutines Class and Group.

Step 1. For each group $h = 1, \dots, H$, do the following:

a. Form n_h batches of jobs, denoted as $B_{1h}, \dots, B_{n_h, h}$, such that each batch consists of exactly s jobs, one from each job class Q_{ih} in G_h for $i = 1, \dots, s$.

b. For $k = 1, \dots, n_h$, schedule the s jobs of batch B_{kh} consecutively in arbitrary order.

Step 2. Schedule the jobs in $N \setminus (\bigcup_{h=1}^H \bigcup_{k=1}^{n_h} B_{kh})$ in arbitrary order at the end.

Steps 1 and 2 each require $O(n)$ time. Thus, the overall time requirement of Heuristic Batch is $O(\max\{sK^2, n\})$. The job sequence generated by Heuristic Batch is

$$B_{11}, \dots, B_{n_1, 1}, B_{12}, \dots, B_{n_2, 2}, \dots,$$

$$B_{1H}, \dots, B_{n_H, H}, N \setminus \left(\bigcup_{h=1}^H \bigcup_{k=1}^{n_h} B_{kh} \right).$$

For each batch B_{kh} , because the parameter vector of each of the s jobs in B_{kh} is a shifted vector of that of each

other job, the total processing time of the s jobs of B_{kh} at each supplier S_i is the same and is equal to P_h , i.e., $\sum_{j \in B_{kh}} p_{ij} = P_h$ for $i = 1, \dots, s$. We begin our analysis of the performance of Heuristic Batch with two preliminary results.

LEMMA 2. *The total completion time, Z^{Batch} , of the schedule generated by Heuristic Batch satisfies*

$$Z^{\text{Batch}} \leq \sum_{h=2}^H s n_h \left(\sum_{j=1}^{h-1} n_j P_j \right) + \sum_{h=1}^H P_h s n_h (n_h + 1)/2 + \left(\sum_{h=1}^H \delta_h \right) \left(\sum_{h=1}^H (n_h + \delta_h) P_h \right).$$

LEMMA 3. *The total completion time, Z^{OPT} , of an optimal schedule for problem $As \parallel \sum C_j$ satisfies*

$$Z^{\text{OPT}} \geq \frac{1}{s} \left[\sum_{h=2}^H g_h \left(\sum_{j=1}^{h-1} g_j P_j \right) + \sum_{h=1}^H P_h (g_h^2 + g_h)/2 \right].$$

THEOREM 14. *If the processing times are generated according to Assumption A, then Heuristic Batch provides solutions that are asymptotically optimal in the number of jobs for problem $As \parallel \sum C_j$. The time complexity of Batch is $O(\max\{sK^2, n\})$.*

PROOF. First, under Assumption A, for each $h = 1, \dots, H$, there exists a constant θ_h with $0 < \theta_h < 1$ and $\sum_{h=1}^H \theta_h = 1$ such that $\lim_{n \rightarrow \infty} n_{ih}/n \stackrel{a.s.}{=} \theta_h$ for each $i = 1, \dots, s$, and that $\lim_{n \rightarrow \infty} n_h/n \stackrel{a.s.}{=} \theta_h$, $\lim_{n \rightarrow \infty} g_h/n \stackrel{a.s.}{=} s\theta_h$, and $\lim_{n \rightarrow \infty} \delta_h/n \stackrel{a.s.}{=} 0$.

It is easy to see that the total completion time of the last batch of jobs in the schedule constructed by Heuristic Batch satisfies

$$\lim_{n \rightarrow \infty} \left(\sum_{h=1}^H \delta_h \right) \left(\sum_{h=1}^H (n_h + \delta_h) P_h \right) / n^2 = 0.$$

Thus, from Lemmas 2 and 3, respectively, we have

$$\lim_{n \rightarrow \infty} \frac{Z^{\text{Batch}}}{n^2} \leq \sum_{h=2}^H s \theta_h \left(\sum_{j=1}^{h-1} \theta_j P_j \right) + \sum_{h=1}^H s P_h \theta_h^2 / 2 \quad \text{and} \\ \lim_{n \rightarrow \infty} \frac{Z^{\text{OPT}}}{n^2} \geq \sum_{h=2}^H s \theta_h \left(\sum_{j=1}^{h-1} \theta_j P_j \right) + \sum_{h=1}^H s P_h \theta_h^2 / 2.$$

Because the right-hand sides of the above two inequalities are finite, these inequalities imply that

$$\lim_{n \rightarrow \infty} \frac{Z^{\text{Batch}} - Z^{\text{OPT}}}{n^2} \stackrel{a.s.}{=} 0 \quad \text{and} \\ \lim_{n \rightarrow \infty} \frac{Z^{\text{Batch}} - Z^{\text{OPT}}}{Z^{\text{OPT}}} \stackrel{a.s.}{=} 0.$$

The time requirement of Step 0 is determined by the running time of Subroutines Class and Group, as discussed

above. Steps 1 and 2 each require $O(n)$ time. Therefore, the overall computation time of Heuristic Batch is $O(\max\{sK^2, n\})$. $\square$

Theorem 14 shows that, although problem $As \parallel \sum C_j$ is formally intractable, close to optimal schedules can be found efficiently for large instances.

The other manufacturer's problem we consider is $As \parallel L_{\max}$. Recall that Theorem 6 shows that this problem can be solved in $O(n \log n)$ time by sequencing the jobs in nondecreasing due date order.

Having studied the schedules which the manufacturer may choose, we now consider the solvability of the resulting suppliers' problems. Here, we first consider problems where the suppliers face due dates implied by the manufacturer's choice of γ . In this case, it is natural for the suppliers to minimize their total cost, subject to a constraint $L_{\max} \leq L$ on the maximum lateness with respect to those due dates. We show that this suppliers' problem is efficiently solvable.

THEOREM 15. *Problem $As | L_{\max} \leq L | \sum \sum C_{ij}$ is solvable in $O(sn \log n)$ time.*

As discussed above, an alternative interpretation of the requirement implied by the manufacturer's choice of γ is that an upper bound constraint $\sum C_j \leq C$ is imposed on the manufacturer's cost.

THEOREM 16. *The recognition version of problem $As | \sum C_j \leq C | \sum \sum C_{ij}$ is unary NP-complete.*

Next, we consider the suppliers' scheduling problem implied by the second requirement to process all jobs in a common sequence, or problem $As | coseq | \sum \sum C_{ij}$. Recall that Theorem 4 shows that this problem can be solved in $O(n \log n)$ time.

Finally, we consider problems where both requirements are imposed by the manufacturer on the suppliers. That is, they need to take γ into account and also process the jobs in a common sequence.

THEOREM 17. *The recognition version of problem $As | coseq, L_{\max} \leq L | \sum \sum C_{ij}$ is unary NP-complete.*

THEOREM 18. *The recognition version of problem $As | coseq, \sum C_j \leq C | \sum \sum C_{ij}$ is unary NP-complete.*

7. Suppliers and Manufacturer Cooperate

Once again motivated by Figure 1, we discuss conflict and cooperation where the suppliers and the manufacturer are all subsidized or fully owned by a single company. The company would like the suppliers and the manufacturer to use the system optimal schedule π because it minimizes the company's total cost. However, because π is in general not optimal for either the suppliers' or the manufacturer's problem, the company may have to provide an incentive for them to use π . We propose one possible incentive scheme that the company can use. Without cooperation between

the suppliers and the manufacturer, either schedule ν or schedule γ will be implemented, depending on whether the suppliers or the manufacturer selects the schedule. Thus, the potential reduction in total cost due to cooperation is at least

$$R_c = \min \left\{ \alpha \sum_{i=1}^s [F_i(\nu) - F_i(\pi)] + (1 - \alpha)[G(\nu) - G(\pi)], \alpha \sum_{i=1}^s [F_i(\gamma) - F_i(\pi)] + (1 - \alpha)[G(\gamma) - G(\pi)] \right\}. \quad (28)$$

The suppliers and the manufacturer can be compensated using the cost saving R_c given in (28). Let u_i and v denote the amount of compensation provided to the supplier S_i for $i = 1, \dots, s$, and the manufacturer, respectively. Because the compensation is funded by the total cost saving, we require

$$\sum_{i=1}^s u_i + v \leq R_c. \quad (29)$$

From Nash (1950, 1953), for the suppliers and the manufacturer to adopt schedule π , the compensation they receive, $(u_1, \dots, u_s, v)$, must make them better off compared to their own optimal schedules. In other words, $(u_1, \dots, u_s, v)$ must satisfy

$$\alpha F_i(\pi) - u_i \leq \alpha F_i(\nu) \quad \text{for } i = 1, \dots, s \quad \text{and} \quad (30)$$

$$(1 - \alpha)G(\pi) - v \leq (1 - \alpha)G(\gamma). \quad (31)$$

In general, if there exists a solution $(u_1, \dots, u_s, v)$ satisfying (29), (30), and (31), then this mechanism can be used to achieve cooperation. This discussion leads to the following result.

THEOREM 19. *There exists a solution $(u_1, \dots, u_s, v)$ satisfying (29), (30), and (31) if and only if schedules ν , γ , and π satisfy the following conditions:*

$$(1 - \alpha)[G(\nu) - G(\pi)] \geq 2\alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] + (1 - \alpha)[G(\pi) - G(\gamma)], \quad (32)$$

$$\alpha \sum_{i=1}^s [F_i(\gamma) - F_i(\pi)] \geq 2(1 - \alpha)[G(\pi) - G(\gamma)] + \alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)]. \quad (33)$$

PROOF. Let

$$u_i = \alpha[F_i(\pi) - F_i(\nu)] \quad \text{for } i = 1, \dots, s \quad \text{and} \quad (34)$$

$$v = (1 - \alpha)[G(\pi) - G(\gamma)]. \quad (35)$$

Clearly, the solution $(u_1, \dots, u_s, v)$ satisfies (30) and (31). Because $\sum_{i=1}^s u_i + v = \alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] + (1 - \alpha)[G(\pi) - G(\gamma)]$, Equations (32) and (33) imply, respectively,

$$\sum_{i=1}^s u_i + v \leq \alpha \sum_{i=1}^s [F_i(\nu) - F_i(\pi)] + (1 - \alpha)[G(\nu) - G(\pi)] \quad \text{and} \quad (36)$$

$$\sum_{i=1}^s u_i + v \leq \alpha \sum_{i=1}^s [F_i(\gamma) - F_i(\pi)] + (1 - \alpha)[G(\gamma) - G(\pi)]. \quad (37)$$

It follows from (36) and (37) that the solution $(u_1, \dots, u_s, v)$ satisfies (29).

($\Leftarrow$) Given a solution $(u_1, \dots, u_s, v)$ satisfying (29), (30), and (31), we have

$$\alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] + (1 - \alpha)[G(\pi) - G(\gamma)] \leq \sum_{i=1}^s u_i + v \leq R_c,$$

which, using (28), implies

$$\alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] + (1 - \alpha)[G(\pi) - G(\gamma)] \leq \alpha \sum_{i=1}^s [F_i(\nu) - F_i(\pi)] + (1 - \alpha)[G(\nu) - G(\pi)] \quad \text{and}$$

$$\alpha \sum_{i=1}^s [F_i(\pi) - F_i(\nu)] + (1 - \alpha)[G(\pi) - G(\gamma)] \leq \alpha \sum_{i=1}^s [F_i(\gamma) - F_i(\pi)] + (1 - \alpha)[G(\gamma) - G(\pi)],$$

which further imply (32) and (33), respectively. $\square$

Theorem 19 shows that if schedules ν , γ , and π satisfy (32) and (33), then the proposed cooperation mechanism can be implemented. Checking these conditions requires finding ν , γ , and π . Results for the suppliers' problems to obtain ν can be found in Theorems 1, 4, 15, 16, 17, and 18. Results for the manufacturer's problems, to obtain γ , can be found in Theorems 2, 6, and 14. Results for the system problems, to obtain π , can be found in Theorems 10, 11, and 12.

Some interpretation of (32) and (33) is now useful. Note that $F_i(\pi) \geq F_i(\nu)$ for $i = 1, \dots, s$, and $G(\pi) \geq G(\gamma)$. Condition (32) means that for the proposed cooperation mechanism to work, the maximum possible cost increase to the manufacturer as a result of no cooperation must be at least twice the total cost increase to all the suppliers plus the cost increase to the manufacturer as a result of cooperation. Similarly, condition (33) means that the maximum possible total cost increase to the suppliers as a result of no cooperation must be at least twice the cost increase to the

manufacturer plus the total cost increase to the suppliers as a result of cooperation.

We now show how the solutions to the suppliers' and manufacturer's problems can be used to check whether the conditions of Theorem 19 can be satisfied, even if π is unknown. In general, the suppliers' problem $As \parallel \sum_{i=1}^s F_i$ and the manufacturer's problem $As \parallel G$ are easier to solve than the system problem $As \parallel \alpha \sum_{i=1}^s F_i + (1 - \alpha)G$. Assuming that we already know ν and γ but not π , we can find an upper bound on the cost of schedule π for the system problem as follows. Condition (32) implies that

$$\begin{aligned} \alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \\ \leq \alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)[G(\nu) + G(\gamma)]/2. \end{aligned} \quad (38)$$

Similarly, condition (33) implies that

$$\begin{aligned} \alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \\ \leq \alpha \sum_{i=1}^s [F_i(\nu) + F_i(\gamma)]/2 + (1 - \alpha)G(\gamma). \end{aligned} \quad (39)$$

Combining (38) and (39), we obtain an upper bound on $\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi)$ given by

$$\begin{aligned} UB(\pi) = \alpha \sum_{i=1}^s F_i(\nu)/2 + (1 - \alpha)G(\gamma)/2 \\ + \min \left\{ \alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)G(\nu), \right. \\ \left. \alpha \sum_{i=1}^s F_i(\gamma) + (1 - \alpha)G(\gamma) \right\} / 2. \end{aligned} \quad (40)$$

Suppose that we also have found a lower bound, denoted by $LB(\pi)$, on $\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi)$, by using mathematical programming techniques. Then, using (40), if $LB(\pi) > UB(\pi)$, we do not need to know π exactly to establish that the proposed cooperation mechanism is not implementable.

8. Cost Saving from Cooperation

We now computationally evaluate the cost saving from the various mechanisms for cooperation between the suppliers and the manufacturer proposed in §§4, 5, and 7.

First, where the suppliers have dominant power in the supply chain, the mechanism for cooperation proposed in §4 provides a net saving given by (22) to the manufacturer. Without cooperation the suppliers impose schedule ν , therefore the relative cost saving to the manufacturer is

$$\begin{aligned} z_{\text{man}} = \left(\left[\alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)G(\nu) \right] \right. \\ \left. - \left[\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \right] \right) \\ \cdot ((1 - \alpha)G(\nu))^{-1}. \end{aligned} \quad (41)$$

Second, where the manufacturer has dominant power in the supply chain, the mechanism for cooperation proposed in §5 provides a total net saving given by (27) to the suppliers. Without cooperation the manufacturer imposes schedule γ , therefore the relative cost saving to the suppliers is

$$\begin{aligned} z_{\text{sup}} = \left(\left[\alpha \sum_{i=1}^s F_i(\gamma) + (1 - \alpha)G(\gamma) \right] \right. \\ \left. - \left[\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \right] \right) \\ \cdot \left(\alpha \sum_{i=1}^s F_i(\gamma) \right)^{-1}. \end{aligned} \quad (42)$$

Finally, where there is a decision maker in the supply chain who makes centralized decisions for the system, the mechanism for cooperation proposed in §7 provides a net saving given by (28) to the system. There are two cases to be considered.

Case 1. If schedule ν is used without cooperation, then the relative cost saving to the system is

$$\begin{aligned} z_{\text{sys}(\nu)} = \left(\left[\alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)G(\nu) \right] \right. \\ \left. - \left[\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \right] \right) \\ \cdot \left(\alpha \sum_{i=1}^s F_i(\nu) + (1 - \alpha)G(\nu) \right)^{-1}. \end{aligned} \quad (43)$$

Case 2. If schedule γ is used without cooperation, then the relative cost saving to the system is

$$\begin{aligned} z_{\text{sys}(\gamma)} = \left(\left[\alpha \sum_{i=1}^s F_i(\gamma) + (1 - \alpha)G(\gamma) \right] \right. \\ \left. - \left[\alpha \sum_{i=1}^s F_i(\pi) + (1 - \alpha)G(\pi) \right] \right) \\ \cdot \left(\alpha \sum_{i=1}^s F_i(\gamma) + (1 - \alpha)G(\gamma) \right)^{-1}. \end{aligned} \quad (44)$$

As throughout the paper, we consider two combinations of cost functions: (1) the suppliers' cost function $F_i = \sum_{j \in N} C_{ij}$ for $i = 1, \dots, s$, and the manufacturer's cost function $G = \sum_{j \in N} C_j$; (2) the suppliers' cost function $F_i = \sum_{j \in N} C_{ij}$ for $i = 1, \dots, s$, and the manufacturer's cost function $G = L_{\max}$. In the following subsections, we estimate the four relative cost savings defined by (41), (42), (43), and (44).

8.1. Completion Time and Completion Time

From Theorem 1, schedule ν consists of an SPT order of the jobs at each supplier. However, from Theorems 2 and 10, finding schedules γ and π is intractable. Thus, instead of finding γ and π exactly, we compute an

upper or lower bound on the cost of an optimal schedule. We derive an upper bound $UB_{\text{sys}}(\pi)$ on the system cost $\alpha \sum \sum C_{ij}(\pi) + (1 - \alpha) \sum C_j(\pi)$, a lower bound $LB_{\text{sup}}(\gamma)$ on the suppliers' total cost $\alpha \sum \sum C_{ij}(\gamma)$, and a lower bound $LB_{\text{man}}(\gamma)$ on the manufacturer's cost $(1 - \alpha) \sum C_j(\gamma)$.

The upper bound $UB_{\text{sys}}(\pi)$ is found using a simulated annealing heuristic (Koulamas et al. 1994) for problem $As\| \alpha \sum \sum C_{ij} + (1 - \alpha) \sum C_j$. An initial schedule is given by an SPT order of the jobs at each supplier. The neighborhood used is pairwise exchange of job positions at a supplier. Any schedule σ that is found becomes the new seed schedule with probability $e^{-a\Delta(\sigma)}$, where $\Delta(\sigma)$ is the relative difference in the costs of σ and the incumbent schedule, and a is a positive parameter. The value of a increases from 10 to 50 during the procedure. If no new incumbent schedule is found during 10,000 consecutive iterations, then the heuristic terminates.

We let $LB_{\text{sup}}(\gamma) = \alpha \sum \sum C_{ij}(\rho)$, where schedule ρ is generated by Heuristic STPT described in §3.1. From (13), $LB_{\text{sup}}(\gamma)$ provides a lower bound on $\alpha \sum \sum C_{ij}(\gamma)$.

The lower bound $LB_{\text{man}}(\gamma)$ is derived from the following observations. It is easy to show that for any given parameters $(v_1, \dots, v_s)$ with $v_i \geq 0$ and $\sum_{i=1}^s v_i = 1$, we have

$$\sum_{j \in N} C_j(\gamma) = \sum_{j \in N} \max_{i=1, \dots, s} \{C_{ij}(\gamma)\} \geq \sum_{j \in N} \sum_{i=1}^s v_i C_{ij}(\gamma). \quad (45)$$

It can be shown that the term $\sum_{j \in N} \sum_{i=1}^s v_i C_{ij}$ is minimized over schedules with identical sequences at all suppliers if the jobs are scheduled in nondecreasing order of $\sum_{i=1}^s v_i p_{ij}$ at all suppliers. Let the schedule corresponding to nondecreasing order of $\sum_{i=1}^s v_i p_{ij}$ be denoted by $\theta(v_1, \dots, v_s)$. From Lemma 1, we may assume that schedule γ uses identical sequences at all suppliers. Therefore,

$$\sum_{j \in N} \sum_{i=1}^s v_i C_{ij}(\gamma) \geq \sum_{j \in N} \sum_{i=1}^s v_i C_{ij}(\theta(v_1, \dots, v_s)). \quad (46)$$

Then, for any nonnegative numbers $v_1, \dots, v_s$ with $\sum_{i=1}^s v_i = 1$, (45) and (46) imply that

$$\sum_{j \in N} C_j(\gamma) \geq \sum_{j \in N} \sum_{i=1}^s v_i C_{ij}(\theta(v_1, \dots, v_s)). \quad (47)$$

We use a simulated annealing heuristic to search for values of $v_1, \dots, v_s$ such that the right-hand side of (47) is as large as possible. This heuristic is similar to that used to find $UB_{\text{sys}}(\pi)$. Let the values of $v_1, \dots, v_s$ found by this heuristic be denoted by $\bar{v}_1, \dots, \bar{v}_s$, respectively. Let $LB_{\text{man}}(\gamma) = (1 - \alpha) \sum_{j \in N} \sum_{i=1}^s \bar{v}_i C_{ij}(\theta(\bar{v}_1, \dots, \bar{v}_s))$. It follows from (47) that $LB_{\text{man}}(\gamma)$ is a lower bound on $(1 - \alpha) \sum_{j \in N} C_j(\gamma)$.

Using the bounds $UB_{\text{sys}}(\pi)$, $LB_{\text{sup}}(\gamma)$, and $LB_{\text{man}}(\gamma)$ to replace, respectively, $\alpha \sum \sum C_{ij}(\pi) + (1 - \alpha) \sum C_j(\pi)$, $\alpha \sum \sum C_{ij}(\gamma)$, and $(1 - \alpha) \sum C_j(\gamma)$ in (41)–(44), we obtain

$$z_{\text{man}} \geq \max \left\{ 0, \frac{\alpha \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu) + (1 - \alpha) \sum_{j \in N} C_j(\nu) - UB_{\text{sys}}(\pi)}{(1 - \alpha) \sum_{j \in N} C_j(\nu)} \right\}, \quad (48)$$

$$z_{\text{sup}} \geq \max \left\{ 0, 1 - \frac{UB_{\text{sys}}(\pi) - LB_{\text{man}}(\gamma)}{LB_{\text{sup}}(\gamma)} \right\}, \quad (49)$$

$$z_{\text{sys}(\nu)} \geq \max \left\{ 0, 1 - \frac{UB_{\text{sys}}(\pi)}{\alpha \sum_{i=1}^s \sum_{j \in N} C_{ij}(\nu) + (1 - \alpha) \sum_{j \in N} C_j(\nu)} \right\}, \quad (50)$$

and

$$z_{\text{sys}(\gamma)} \geq \max \left\{ 0, 1 - \frac{UB_{\text{sys}}(\pi)}{LB_{\text{sup}}(\gamma) + LB_{\text{man}}(\gamma)} \right\}. \quad (51)$$

We compute the lower bounds in (48)–(51) using random test problems generated as follows: (i) the number of suppliers $s \in \{2, 4, 8\}$; (ii) the number of jobs $n \in \{25, 50, 100\}$; (iii) the job processing times p_{ij} are independently drawn from a uniform integer distribution over the interval $[10, 100]$; and (iv) the relative cost parameter $\alpha \in \{0.2, 0.5, 0.8\}$. For each (s, n, α) combination, 10 random instances are generated and the lower bounds on z_{man} , z_{sup} , $z_{\text{sys}(\nu)}$, and $z_{\text{sys}(\gamma)}$ are computed. The mean lower bounds over the 10 random instances are reported in Table 5(a). The cost saving depends on the relative cost parameter (α) and the problem size (s, n) . The relative cost saving to the manufacturer (z_{man}) and to the system ($z_{\text{sys}(\nu)}$) when the suppliers' optimal schedule ν is used without cooperation is larger when α is small. In particular, z_{man} and $z_{\text{sys}(\nu)}$ are always at least 12.5% when $\alpha \leq 0.2$. By contrast, the relative cost saving to the suppliers as a whole (z_{sup}) and to the system ($z_{\text{sys}(\gamma)}$) when the manufacturer's optimal schedule γ is used without cooperation is larger when α is large. In particular, z_{sup} and $z_{\text{sys}(\gamma)}$ are always at least 10.1% when $\alpha \geq 0.5$ and $s \geq 8$. The relative cost savings on z_{man} (respectively, z_{sup}) are consistently larger than those on $z_{\text{sys}(\nu)}$ (respectively, $z_{\text{sys}(\gamma)}$). The reason is that a schedule typically has more impact on an individual party in the system (i.e., the suppliers or the manufacturer) than on the entire system. This is because positive and negative effects may cancel each other out within the system objective, hence the overall system performance may not be as good (or as bad).

8.2. Completion Time and Maximum Lateness

From Theorem 1, schedule ν is an SPT order of the jobs at each supplier. From Theorem 6, schedule γ is an EDD order of the jobs. Schedule π , which is optimal for problem $As\| \alpha \sum \sum C_{ij} + (1 - \alpha) L_{\text{max}}$, can be found by Algorithm SCL in §4 which from Theorem 12 runs in polynomial time.

We compute the relative cost savings z_{man} , z_{sup} , $z_{\text{sys}(\nu)}$, and $z_{\text{sys}(\gamma)}$ defined, respectively, in (41), (42), (43), and (44) using random test problems generated as follows: (i) s , n , and p_{ij} are generated as in §8.1; (ii) the job due dates d_j are independently drawn from a uniform integer distribution over the interval $[10, 55n]$; and (iii) the relative cost parameter $\alpha = \beta/n$, where $\beta \in \{0.5, 2, 8\}$. The values of α

Table 5. Relative cost saving from cooperation.

s	n	α	Lower bound on relative cost saving			
			z_{man} (%)	z_{sup} (%)	$z_{\text{sys}(v)}$ (%)	$z_{\text{sys}(\gamma)}$ (%)
(a) Completion time and completion time						
2	25	0.2	16.76	0.00	12.50	0.00
		0.5	10.93	1.66	4.63	1.10
		0.8	4.30	6.45	0.67	5.72
2	50	0.2	18.58	0.00	13.76	0.00
		0.5	12.15	1.53	5.08	1.02
		0.8	4.62	5.82	0.70	5.17
2	100	0.2	18.51	0.00	13.79	0.00
		0.5	11.54	1.99	4.87	1.33
		0.8	3.75	6.71	0.58	5.96
4	25	0.2	23.95	0.00	15.57	0.00
		0.5	14.99	6.17	4.75	4.90
		0.8	6.07	11.33	0.63	10.64
4	50	0.2	25.89	0.03	16.88	0.01
		0.5	16.06	6.88	5.12	5.49
		0.8	5.95	11.92	0.62	11.21
4	100	0.2	26.31	0.00	17.17	0.00
		0.5	15.93	7.01	5.09	5.60
		0.8	4.28	12.04	0.45	11.32
8	25	0.2	24.94	2.66	13.07	1.73
		0.5	15.08	11.47	3.25	10.12
		0.8	6.04	15.41	0.39	14.91
8	50	0.2	26.76	3.71	14.05	2.45
		0.5	16.07	12.31	3.48	10.91
		0.8	4.59	16.23	0.3	15.72
8	100	0.2	27.02	3.49	14.24	2.32
		0.5	14.28	12.16	3.11	10.80
		0.8	0.13	16.41	0.01	15.91
(b) Completion time and maximum lateness						
s	n	α	Relative cost saving			
			z_{man} (%)	z_{sup} (%)	$z_{\text{sys}(v)}$ (%)	$z_{\text{sys}(\gamma)}$ (%)
2	25	0.5/25	69.85	12.68	47.00	9.29
		2/25	56.12	16.50	18.35	15.22
		8/25	34.67	21.34	2.87	21.02
2	50	0.5/50	78.35	13.46	54.74	10.71
		2/50	66.35	17.58	23.90	16.53
		8/50	46.07	21.94	5.06	21.64
2	100	0.5/100	82.37	12.70	58.38	10.74
		2/100	69.26	17.14	25.97	16.40
		8/100	50.01	22.13	6.18	21.90
4	25	0.5/25	64.00	16.76	33.89	13.63
		2/25	49.63	19.70	10.83	18.69
		8/25	28.18	22.71	1.32	22.48
4	50	0.5/50	74.85	16.86	41.24	14.73
		2/50	60.18	20.23	13.81	19.55
		8/50	39.81	23.45	2.44	23.27
4	100	0.5/100	77.63	15.76	43.37	14.19
		2/100	62.38	19.88	14.86	19.36
		8/100	43.73	23.45	3.00	23.31
8	25	0.5/25	60.68	19.17	22.54	17.08
		2/25	44.25	21.52	5.40	20.91
		8/25	24.01	23.55	0.60	23.42
8	50	0.5/50	68.37	19.00	26.59	17.50
		2/50	51.56	21.93	6.91	21.49
		8/50	32.76	24.18	1.08	24.07
8	100	0.5/100	71.17	18.51	28.14	17.43
		2/100	54.05	22.31	7.51	21.98
		8/100	36.69	24.84	1.34	24.75

are selected such that both cost components play a significant role and that a wide range of values for the relative cost saving is considered. Because the completion time of the last job is approximately $55n$ and the due dates are generated from the interval $[10, 55n]$, it can be expected that L_{max} should be $O(n)$ almost surely. Because $\sum \sum C_{ij}$ is $O(n^2)$, by using $\alpha = O(1/n)$ in our experiment, the magnitude of $\alpha \sum \sum C_{ij}$ matches that of $(1 - \alpha)L_{\text{max}}$ at $O(n)$, which makes our test problems representative. For each (s, n, α) combination, 10 random instances are generated and the values of z_{man} , z_{sup} , $z_{\text{sys}(\nu)}$, and $z_{\text{sys}(\gamma)}$ are computed. The mean values of these measures over the 10 random instances are reported in Table 5(b). From these results we conclude that, in most cases, the proposed cooperation mechanism results in significant cost reductions for the decision makers. In fact, all results for z_{man} , z_{sup} and $z_{\text{sys}(\gamma)}$ show a relative cost saving of at least 9.2%. The explanation of the results follows that for Table 5(a).

9. Extensions

In this section, we consider two practical extensions of our assembly system model. Section 9.1 considers nonzero transportation times between the suppliers and the manufacturer. Section 9.2 considers bottleneck processing operations at the manufacturer.

REMARK 1. All the cooperation mechanisms described in §§4 through 7 can be used exactly as described there, in the presence of transportation times and bottleneck operations at the manufacturer.

As a consequence of Remark 1, this section focuses on differences in the solvability of the manufacturer's or system problem which arise from transportation times or bottleneck manufacturer operations.

9.1. Transportation Times

Here we consider the solvability of scheduling problems in assembly systems with arbitrary transportation time t_{ij} for moving parts for job j from supplier S_i to the manufacturer for $i = 1, \dots, s$, $j = 1, \dots, n$. From Theorem 2, we already know that the recognition version of problem $As|t_{ij}| \sum C_j$ is unary NP-complete. Therefore, we only consider the solvability of problems with the L_{max} objective. First, we show by the following example that Lemma 1 does not hold in the presence of transportation times.

EXAMPLE 1. Consider the following instance of problem $As|t_{ij}|L_{\text{max}}$: $s = 2$; $p_{11} = p_{12} = p_{21} = p_{22} = 1$; $t_{11} = 2$, $t_{12} = 1$, $t_{21} = 1$, $t_{22} = 2$; $d_1 = d_2 = 3$. Now, consider the schedule $\{1, 2\}$ at supplier S_1 and $\{2, 1\}$ at supplier S_2 . In this schedule, $C_1 = d_1 = 3$, $C_2 = d_2 = 3$, and thus $L_{\text{max}} = 0$. Because this is the only schedule with $L_{\text{max}} \leq 0$, Lemma 1 does not hold.

However, the result in Example 1 does not prevent the efficient solution of three related scheduling problems.

First, recall that Theorem 6 shows that problem $As||L_{\max}$ can be solved in $O(n \log n)$ time. We need to study the generalization of this problem which includes transportation times.

THEOREM 20. *Problem $As|t_{ij}|L_{\max}$ is solvable in $O(ns \log n)$ time.*

Second, in the models discussed in §§4 and 5, a system optimal schedule π is always achievable. Therefore, we need to study problem $As|t_{ij}|\alpha \sum \sum C_{ij} + (1 - \alpha)L_{\max}$, which generalizes the problem considered in Theorem 12.

THEOREM 21. *Problem $As|t_{ij}|\alpha \sum \sum C_{ij} + (1 - \alpha)L_{\max}$ is solvable in $O(sn^3 \log n)$ time.*

Finally, we consider the model in §6, where the dominant manufacturer determines a schedule to which the suppliers adjust. This suggests the need to study problem $As|t_{ij}, L_{\max} \leq L|\sum \sum C_{ij}$, which generalizes the problem considered in Theorem 15.

THEOREM 22. *Problem $As|t_{ij}, L_{\max} \leq L|\sum \sum C_{ij}$ is solvable in $O(ns \log n)$ time.*

9.2. Bottleneck Operations at the Manufacturer

In the presence of bottleneck operations at the manufacturer, we are only able to provide negative results about the solvability of scheduling problems in assembly systems. The recognition versions of the manufacturer's or system problems with $\sum C_j$ objective are unary NP-complete because they are generalizations of problem $As||\sum C_j$, as discussed in Theorem 2. The recognition versions of the manufacturer's and system problems with $L_{\max}$ objective are unary NP-complete because they are generalizations of the two-machine flow-shop problem $F2||L_{\max}$ (Lenstra et al. 1977).

10. Concluding Remarks

In this paper, we first identify and evaluate the substantial costs which arise from conflicts between the respective optimal schedules of the suppliers and manufacturer in an assembly system. Then, we describe several cooperation mechanisms by which the suppliers and manufacturer can resolve this conflict, depending upon their relative bargaining power. Evaluating the cost of conflict and the benefit of cooperation requires various scheduling problems to be solved by the suppliers, the manufacturer, and the overall system. For all the scheduling problems which arise in this way, we provide either an efficient algorithm or a proof of intractability. For two of these scheduling problems that we show are intractable, we also describe heuristics and analyze their worst case performance or demonstrate asymptotic optimality of their solutions. Finally, extensions of our models to consider complications caused by bottleneck operations at the manufacturer and by nonzero transportation times are discussed.

Several important research issues remain open for future investigation. First, while we illustrate how to design and analyze heuristics for two intractable problems that arise in assembly systems, there are several other problems where a similar analysis is needed. A second important research topic is to extend our cooperation mechanisms to multi-stage assembly systems, where relatively few such problems are likely to be tractable. Another possible topic is the modelling of cooperation between the suppliers. We also encourage the development of models of the bargaining process between suppliers and a manufacturer that consider practical issues such as imperfect or asymmetric information. Another significant issue is how, if necessary, to enforce cooperation between the parties. Related to this, studies that examine the benefits of cooperation in supply chains should also consider the value of organizational integration because this makes cooperation easier to establish and maintain.

In several other manufacturing environments, besides those considered here and in Hall and Potts (2003), Dawande et al. (2006), and Agnetis et al. (2006), potentially significant cost savings motivate research into supply chain scheduling issues. These four papers illustrate the generality and importance of cooperation in scheduling decisions in supply chains. As the substantial supply chain management literature attests, there are also several other types of decisions where cooperation can be valuable. However, the benefits there are often not as easy to identify and evaluate as they are for scheduling decisions. This observation emphasizes the value of scheduling models within the supply chain management literature. In conclusion, we hope that our work will focus the attention of supply chain management researchers more toward operational issues in deterministic environments because this is where research is currently most needed and also where optimization models are most effective.

11. Electronic Companion

An electronic companion to this paper is available as part of the online version that can be found at <http://or.journal.informs.org/>.

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